

**AMENDMENT NO. 2 APRIL 1994
TO
IS 13021 (Part 2) : 1991 ac SUPPLIED ELECTRONIC
BALLASTS FOR TUBULAR FLUORESCENT LAMPS —
SPECIFICATION**

PART 2 PERFORMANCE REQUIREMENTS

(*Second cover page, Foreword, para 1*) — Insert the following additional paras after para 1

The ballasts covered under this standard are intended to operate lamps at various frequencies including high frequencies. Attention is drawn to the fact that operating frequencies below 20 kHz may cause audio noise disturbances whereas frequencies above 50 kHz may increase radio interference problems.

Some lamps may be specifically designed for high frequency operation on high frequency ballasts. The starting modes, preheat and non preheat are described.

NOTE — The possibility exists for operation of lamps designed for perfect starting on circuits of the non preheat type. Lamps specified for operation on both types of circuits may appear in IS 2418 (Part 1) 1977 and IS 2418 (Part 2) 1977 or lamp manufacturer will have to authorize such operation of their lamps.

In order to obtain satisfactory performance of fluorescent lamps and electronic ballast, it is necessary that certain features of their design be properly coordinated. It is essential, therefore, that specifications for these be written in terms of measurement made against some common baseline of reference, which must be reasonable, permanent and reproducible.

These conditions may be fulfilled by reference ballasts. Moreover the testing of ballasts for fluorescent lamps will, in general, be made with reference lamps, and in particular by comparing results obtained on such lamps with ballasts to be tested and with a reference ballast.

Whereas the reference ballast for frequencies of 50 Hz is a self inductive coil, the high frequency reference ballast is a resistor because of the impedance of frequency and the back of influence of parasitic capacitance.

(*Page 1, clause 2.1*) — Inset the following at the end

<i>IS No</i>	<i>Title</i>
10321 (Part 1) 1991	ac Supplied electronic ballast for tubular fluorescent lamps. Part 1 general and safety requirements

(Page 2, clause 5.2 (a) line 2) —Substitute 'or' for 'of'.

(Page 2, clause 6, Title) — Replace the title with 'General Requirements' in place of 'General Statement'

(Page 5, clause 7.3.1, para 1) — Insert the following para after the first para

'For lamps not covered by IS 2418 (Part 2) 1977, the peak value of open circuit voltage shall not exceed 700 V'

(Page 5, clause 7.3.2, para 1) — Insert the following para at the end.

'For lamps not covered by IS 2418 (Part 2) 1977, the value of substitution resistor declared by the manufacturer shall be used'

(Page 5, clause 7.3.3.1) — Insert the following at the end:

'For lamps not covered by IS 2418 (Part 2) 1977, the value of substitution resistor declared by the manufacturer shall be used'

(Page 5, clause 7.3.3.1, line 6, formula) — Substitute ' R_1 ' for ' R_0 '.

(Page 7, clause 15.1, Informal Table) — Insert the following note in this table at the end

'In case of equipment schaffners NSG 223 A the rise time for slow high energy transient type/level may be taken as 500 ns for ASY 3 000 V and 100 ns for SYM 1 000 V'

(Page 7, clause 16.2) — Insert the following para at the end

'For lamps not covered by IS 2418 (Part 2) 1977, the value of dummy cathode resistor declared by the manufacturer shall be used.'

(Page 8, clause 17.2.1) — Insert the following new clause after this clause:

'18 SAFETY REQUIREMENTS

18.1 The ballast shall meet the safety requirements as given in 7 to 19 of IS 13021 (Part 1) . 1991'

(ETD 23)

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